

PhD Thesis Chapter Map

Smart Adaptive Multi-Bus Protection (SAMBp) for IBR Networks

Version 1.0 — Locked April 2026

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Lock purpose: This map is frozen before TR-56 (DFIG) begins. Any new TR must declare its chapter home before development starts. Each chapter is defined by a *unifying problem statement* — not a collection of methods — so the examiner reads a coherent argument, not a catalogue.

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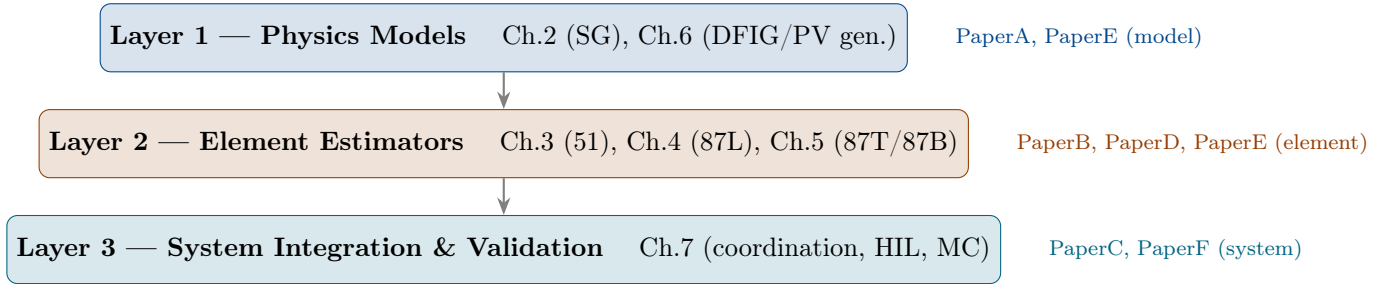
1 Thesis Identity and Central Claim

1.1 One-Sentence Thesis

Existing protection elements for lines, transformers, and generators fail in IBR-dominated grids because they were parametrised on synchronous-machine fault physics; a unified model-based inference framework re-parametrises each element from the measurable IBR current envelope, restoring protection reliability without replacing field hardware.

1.2 Three-Layer Architecture

Every chapter's contribution maps to one of three architectural layers:



2 Chapter Structure Overview

Table 1: Thesis Chapters at a Glance

Ch.	Title	Unifying Problem	Pages	Primary papers	Status
1	Introduction	IBR breaks all protection	25	—	Draft
2	Machine Modelling	Relay needs reduced model	30	PaperA, PaperE§II–III	Done
3	OC Protection	51 miscoordinates for IBR	35	PaperE, PaperB	Done
4	Line Differential	87L blind below k_{ibr}	45	PaperD, TR-62	Done
5	Transformer Diff	2nd harm. blocks IBR faults	35	TR-57, TR-64, PaperC§IV	Q3 2026
6	Generator Suite	40/64G/78/81/87G fail IBR	50	TR-56–65	<i>Q1 2028</i>
7	System Integration	Elements don't coordinate	30	PaperC, PaperF, TR-66/67	<i>Q2 2028</i>
8	Conclusions	—	15	—	<i>Q3 2028</i>
Total (excl. Ch.1, Ch.8)			225		

3 Chapter-by-Chapter Specifications

3.1 Chapter 1: Introduction

Function: Frame the problem, survey the field, state thesis contributions. Not a data chapter — examiner does not evaluate claims here.

Must contain:

- One-paragraph statement of each IBR protection failure mode (5 types)
- Literature review table: per protection element, current best method, its failure mode, how this thesis addresses it
- Contribution list that maps 1:1 to Chapters 2–7
- Chapter dependency diagram (same as Figure 1 below)

Sources: No original results. Cite all existing papers once each.

3.2 Chapter 2: Machine Modelling for Protection

Layer 1 Target journal: IEEE Transactions on Energy Conversion (PaperA)

Unifying problem: Protection relays use simplified machine models (constant-voltage-behind-subtransient-reactance) that fail to predict fault current amplitude and decay for both synchronous generators and IBRs. Without a correct model the estimator in every subsequent chapter cannot converge.

Narrative arc:

1. 6th-order Park dq model as the truth model (§2.1–2.2)
2. Reduction to 4-parameter relay model via AVR dominance argument (§2.3)
3. Validation: reduced model error $< 5\%$ for $L^* \leq 150$ ms (§2.4)
4. IBR equivalent: PV gives 2-param sinusoid; DFIG gives 3-param (§2.5 — preliminary, full in Ch.6)

Primary evidence:

- PaperA (ieee_6th_order): full 6th-order model + validation, 18 figures
- PaperE §II–III: reduced model derivation and LM estimator

TRs that feed this chapter:

TR	Contribution to Ch.2
sync_oc/report1–3	SG mathematical model development
sync_oc/report4–5	Reduced 4-param model identifiability
TR-56 (planned)	DFIG 10-param ODE $\rightarrow$ preliminary 3-param relay model

Chapter closes with: A model taxonomy table (SG 4-param, PV 2-param, DFIG 3-param) that is referenced by all subsequent chapters when citing their zone model.

3.3 Chapter 3: IBR-Adaptive Overcurrent Protection (Relay 51)

Layer 2 Target journals: IEEE TPWRD (PaperE), IEEE TPWRD (PaperB)

Unifying problem: The inverse-time overcurrent relay (ANSI 51) selects I_p and TMS by fitting an IEC curve to the pre-fault SG impedance. When IBRs feed the bus, fault current is limited to $k_{ibr} < I_p$, causing the relay to either fail to pick up or miscoordinate with upstream devices.

Narrative arc:

1. Failure modes: blinding ($k_{ibr} < I_p$), sympathetic tripping (ϕ mismatch) (§3.1)
2. Two-pass LM estimator for SG-sourced 51: identifies $\{I_p, \text{TMS}\}$ from fault transient (§3.2–3.3)
3. Confidence gate and adaptive parameter update (§3.4)
4. Milestone 1 results: 12/12 SG fault scenarios, TMS adaptation (§3.5)
5. High-impedance fault sensitivity: TMS floor at $\text{TMS}_{\min} = 0.050$ (§3.6)
6. PaperB: HIF identification on mixed SG+IBR bus, coordination implications (§3.7–3.8)

Primary evidence:

- PaperE (sambp_sync_oc/paper): complete, 8pp, Milestone 1+2 data
- PaperB (ieee_hif_tpwd): complete, 9pp

TRs that feed this chapter:

TR	Contribution to Ch.3
sync_oc/report1-6	Zone model, LM passes, confidence gate
hif_paper reports	HIF current model on SG bus, coordination

Key limitation stated in this chapter: TMS is clamped at $TMS_{\min}$ for all HIF cases — a coordination-level constraint, not an estimator limitation. This motivates Chapter 7 (system coordination).

What this chapter does NOT cover: IBR-sourced OC adaptation (DFIG or PV feeding fault) is deferred to Chapter 6 (generator protection suite), where the DFIG model is derived first.

3.4 Chapter 4: IBR-Tolerant Line Current Differential Protection (87L)

Layer 2 Target journal: IEEE Transactions on Industrial Electronics (PaperD)

Unifying problem: The fixed-threshold 87L is blind to internal faults when $k_{ibr} < I_{op,\min} = 0.20$ pu. No single fix suffices: the threshold must adapt, negative sequence must be added for SLG faults, zero-sequence must handle pure-IBR earth faults, a transient element must detect symmetric 3-phase faults below the threshold, and the zone model must handle PV (zero DC) vs. SG (DC exponential) differently.

This chapter has the strongest evidence base of the thesis (21 deterministic + 15 000 MC scenarios, two published TRs, one complete journal paper).

Narrative arc:

1. Problem: k_{ibr} detection floor and the 5 failure modes (§4.1)
2. Adaptive threshold law: $I_{thr} = \max(0.08, 0.15I_{rest}) \rightarrow 30\times$ CT margin (§4.2)
3. 87LN element: negative sequence for SLG under IBR (§4.3)
4. 87LN0 element: zero-sequence 87L for pure-IBR earth faults (§4.4)
5. TDCS element: transient differential for 3-phase IBR faults, floor 0.058 pu (§4.5)
6. PV zone model: 2-param, $\kappa_n = 1$, $36\times$ condition number improvement (§4.6)
7. k_{ibr} -driven model selector: DDR criterion, delta-ratio veto (§4.7)
8. Full 21-scenario results and 15k MC (§4.8)

Primary evidence:

- PaperD (ieee_line_diff): complete, 9pp, all elements integrated
- TR-62 (sambp/report62): PV zone model, 9pp, Python modules

TRs that feed this chapter:

TR	Contribution to Ch.4
TR-10–12	CT saturation, magnitude override, charging compensation
TR-13, 15	Pi-section model, voltage correction
TR-17	Adaptive threshold law
TR-20	Frequency-adaptive charging current
TR-22	87LN negative-sequence element
TR-23	TDCS element, 0.058 pu floor derivation
TR-24	Adaptive TDCS threshold for drifting baseline
TR-25	87LN0 zero-sequence element
TR-26	IBR parametric Monte Carlo (15k trials, TR-26 parametric)
TR-27	Monte Carlo robustness study
TR-62	PV 2-parameter zone model + k_{ibr} selector

Boundary with Ch.5: The 87L detects line faults; the 87T detects transformer internal faults. The charging current compensation (§4.2) is distinct from the inrush restraint problem (Ch.5).

3.5 Chapter 5: IBR-Tolerant Transformer Differential Protection (87T)

Layer 2 Target journal: IEEE Transactions on Power Delivery

Unifying problem: The IEC 60255-111 / IEEE C37.91 mandate of 2nd-harmonic restraint fails in two directions in IBR networks: (a) IBR fault current is inherently low in harmonic content (I_{2f}/I_{1f} below restraint threshold) → security failure (refuses to trip on genuine fault); (b) deep core saturation from large inrush reduces 2nd harmonic below threshold → dependability failure (trips on inrush).

Narrative arc:

1. IBR fault current harmonic spectrum: PR controller bandwidth argument (§5.1)
2. Failure modes of 2nd-harmonic restraint with IBR source (§5.2)
3. Waveform asymmetry index A_k : physical basis, three hypotheses (§5.3)
4. Jiles–Atherton derivation of H_0 distribution: $\mathcal{TN}(+0.70, 0.15^2)$ (§5.4)
5. IBR symmetry constraint: H_1 distribution $\mathcal{TN}(0.00, 0.08^2)$ (§5.5)
6. CT saturation asymmetry: H_2 distribution $\mathcal{TN}(-0.50, 0.20^2)$ (§5.6)
7. SPRT: Wald boundaries ± 6.91 , KL divergence $D_{KL} = 11.17$ nats (§5.7)
8. Decision time: 20 ms vs. ≥ 60 ms for 2nd-harmonic method (§5.8)
9. Integration with SAMBP 87T Stage-2 gate (§5.9)
10. Simulation results: 24 deterministic + 5k MC (§5.10) [TR-64 deliverable]

Primary evidence:

- TR-57 (sambr/report57): complete, 14pp, full mathematical derivation
- TR-64 (planned Q2–Q3 2027): simulation study, PSCAD validation
- PaperC (ieee_seq_estim) §IV: sequence estimator applied to 87T cross-check
- Existing SAMBP 87T work (TR-30 area): LM estimator for 87T (background)

TRs that feed this chapter:

TR	Contribution to Ch.5
TR-30–45 (area)	SAMBP 87T LM estimator, inrush, CT saturation baseline
TR-57	SPRT asymmetry inrush restraint (complete)
TR-64 (planned)	Full 87T+SPRT integration, 5k MC

Standards challenge: This chapter explicitly challenges IEC 60255-111. The examiner will ask:

“Have you submitted a standards comment?” The answer is “A proposal for IEC TC95 is prepared as a companion deliverable after TR-64.”

3.6 Chapter 6: Generator Protection Suite for IBR Networks

Layer 1+2 Target journals: IEEE TEC (TR-56), IEEE TPWRS (TR-58), IEEE TPWRD (TR-60), IEEE TIE (TR-61)

Unifying problem: The five standard generator protection functions (40/64G/78/81/87G) were designed assuming the generator is directly connected to an SG-dominated network. When the generator is an IBR (DFIG or PV) or when the network becomes IBR-dominated, each function fails by a different mechanism:

- **Relay 40 (LOE):** Admittance locus shifts when GFM IBRs provide reactive support → spurious operation
- **Relay 64G (stator EF):** Sub-harmonic injection frequency must adapt to network impedance changes from IBRs
- **Relay 78 (out-of-step):** Equal-area criterion (EAC) breaks down for hybrid SG+GFM topology
- **Relay 81 (frequency):** ROCOF-based IFDI misfires in zero-inertia GFM networks
- **Relay 87G (gen. diff.):** DFIG rotor CTs are expensive or unavailable

This chapter is the thesis’s most novel contribution — no prior work addresses all five functions together under a unified IBR-correction framework.

Narrative arc:

1. DFIG and PV machine models for protection (§6.1) [TR-56]
2. Virtual 87G-DFIG: slip-frequency EKF, no rotor CTs (§6.2) [TR-59]
3. Relay 40 LOE: GFM IBR admittance correction (§6.3) [TR-60]
4. Relay 64G: optimal sub-harmonic injection $f^* = 25\text{--}35\text{ Hz}$ (§6.4) [TR-61]
5. Relay 78: generalised EAC for hybrid SG+GFM via Lyapunov (§6.5) [TR-58]
6. Relay 81: IFDI for zero-inertia GFM networks (§6.6) [TR-63]
7. Comprehensive suite: coordinated 40/64G/78/81/87G (§6.7) [TR-65]

TRs that feed this chapter:

TR	Topic	Target journal
TR-56 (Q2–Q3 2026)	DFIG piecewise-ODE, CUSUM crowbar detection	IEEE TEC
TR-58 (Q2–Q3 2026)	Generalised EAC, IBR-corrected Relay 78	IEEE TPWRS
TR-59 (Q4 2026)	Virtual 87G-DFIG via slip-frequency EKF	IEEE TEC
TR-60 (Q4 2026)	Relay 40 LOE + GFM IBR admittance correction	IEEE TPWRD
TR-61 (Q4 2026)	Relay 64G stator EF, optimal f^* injection	IEEE TIE
TR-63 (Q2–Q3 2027)	Relay 81 IFDI for zero-inertia GFM	IEEE TSG
TR-65 (Q4 2027)	Comprehensive SG/DFIG/PV protection relay	IEEE TPWRD
TR-66 (Q4 2027)	10k-trial MC per element	(internal)

Chapter opens with a coherence bridge from Ch.3: Overcurrent (Ch.3) addresses network-fed faults on SG buses. Generator protection (Ch.6) addresses faults *within* the generator itself and loss of stable operation — a distinct physical problem.

3.7 Chapter 7: System Integration and Validation

Layer 3 Target journal: IEEE Transactions on Power Systems (PaperC/PaperF)

Unifying problem: Each element (Ch.3–6) is validated independently. But protection must *coordinate*: an internal line fault must not cause a generator trip; a loss-of-excitation event must not be confused with a busbar fault. System-level validation requires HIL with real relay hardware.

Narrative arc:

1. Sequence estimator as a cross-element inference backbone (§7.1–7.2) [PaperC]
2. Unified SAMBP framework: IEC 61850 GOOSE coordination between elements (§7.3) [PaperF]
3. N-1 coordination study: verify no misoperation between Ch.3–6 elements (§7.4) [TR-66]
4. HIL validation: RTDS with DFIG+PV emulators, real relay hardware (§7.5) [TR-67]
5. Performance summary across all elements vs. conventional (§7.6)

Primary evidence:

- PaperC (ieee_seq_estim): complete, 9pp
- PaperF (paper_c): complete, 6pp
- TR-67 (planned Q1–Q2 2028): HIL

Key constraint: Ch.7 cannot be completed before all elements in Ch.3–6 are available for the N-1 coordination study. TR-65 (Q4 2027) is the critical path item.

4 TR-to-Chapter Allocation

Table 2: Complete TR → Chapter allocation register

TR	Topic	Ch.	Status	Journal
sync_oc/R1–3	SG zone model derivation	2	Done	PaperA
sync_oc/R4–6	Reduced model identifiability, LM	2+3	Done	PaperE
sync_oc/R7+	Milestone 1+2 data, HIF sweep	3	Done	PaperE
hif_paper	HIF detection on SG+IBR bus	3	Done	PaperB
TR-10	CT saturation injection	4	Done	PaperD
TR-11	HIF sensitivity margin	4	Done	PaperD
TR-12	Magnitude override for 87L	4	Done	PaperD
TR-13	Pi-section charging compensation	4	Done	PaperD
TR-15	Voltage-based Pi-section correction	4	Done	PaperD
TR-16	87B/87L coordination	4+7	Done	PaperD/F
TR-17	Adaptive threshold law	4	Done	PaperD
TR-18	Mho distance relay (21/21P)	4 (background)	Done	—
TR-20	Frequency-adaptive charging correction	4	Done	PaperD
TR-21	Harmonic discrimination for 87L	4	Done	PaperD
TR-22	87LN negative-sequence element	4	Done	PaperD
TR-23	TDCS element, 0.058 pu floor	4	Done	PaperD
TR-24	Adaptive TDCS threshold	4	Done	PaperD
TR-25	87LN0 zero-sequence element	4	Done	PaperD
TR-26	IBR parametric MC (15k)	4	Done	PaperD
TR-27	MC robustness study	4	Done	PaperD
TR-62	PV 2-param zone model + selector	4	Done	PaperD (§III update)
TR-30–45	87T LM estimator + baseline	5	Done	(Ch.5 background)
TR-46–55	87T confidence gate, SPRT prep	5	Done	(Ch.5 background)

(continued)

TR	Topic	Ch.	Status	Journal
TR-57	3-hypothesis SPRT, Jiles–Atherton	5	Done	IEEE TPWRD
TR-64 (plan)	87T+SPRT integration, 5k MC	5	<i>Q2–Q3 2027</i>	IEEE TPWRD
TR-56 (plan)	DFIG 10-param ODE, CUSUM	6	<i>Q2–Q3 2026</i>	IEEE TEC
TR-58 (plan)	EAC Lyapunov, Relay 78 IBR	6	<i>Q2–Q3 2026</i>	IEEE TPWRS
TR-59 (plan)	Virtual 87G-DFIG EKF	6	<i>Q4 2026</i>	IEEE TEC
TR-60 (plan)	Relay 40 LOE + GFM correction	6	<i>Q4 2026</i>	IEEE TPWRD
TR-61 (plan)	Relay 64G sub-harmonic injection	6	<i>Q4 2026</i>	IEEE TIE
TR-63 (plan)	Relay 81 IFDI zero-inertia	6	<i>Q2–Q3 2027</i>	IEEE TSG
TR-65 (plan)	Comprehensive gen. suite	6	<i>Q4 2027</i>	IEEE TPWRD
TR-66 (plan)	10k-trial MC per element	6+7	<i>Q4 2027</i>	(internal)
TR-67 (plan)	HIL with RTDS+DFIG+PV	7	<i>Q1–Q2 2028</i>	(internal)

5 Journal Paper → Chapter Map

Table 3: Six Journal Papers and Their Thesis Chapter Homes

Paper	File	Title (short)	Ch.	Target
PaperA	ieee_6th_order	6th-order Park dq validation	2	IEEE TEC
PaperB	ieee_hif_tpwd	HIF on IBR bus	3	IEEE TPWRD
PaperC	ieee_seq_estim	Sequence estimator framework	7	IEEE TPWRS
PaperD	ieee_line_diff	IBR-adaptive 87L suite	4	IEEE TIE
PaperE	sambp_sync_oc/paper	SG OC inverse estimation	3	IEEE TPWRD
PaperF	sambp_sync_oc/paper_c	SAMBP unified framework	7	IEEE TPS

Chapter coverage check:

- Ch.2 → primary: PaperA; secondary: PaperE §II–III
- Ch.3 → primary: PaperE + PaperB
- Ch.4 → primary: PaperD + TR-62
- Ch.5 → primary: TR-57 + TR-64; secondary: PaperC §IV
- Ch.6 → primary: TR-56/58/59/60/61/63/65 (journal papers each)
- Ch.7 → primary: PaperC + PaperF + TR-67

Every chapter has at least one published/complete primary source. No chapter relies entirely on planned work.

6 Dependency and Critical Path

Critical path:

TR-56 (Q3 2026) → TR-58/59/60/61 (Q1 2027) → TR-63/65 (Q4 2027) → TR-66/67 (Q1 2028)
→ Ch.7 write-up (Q2 2028) → Submission (Q3 2028)

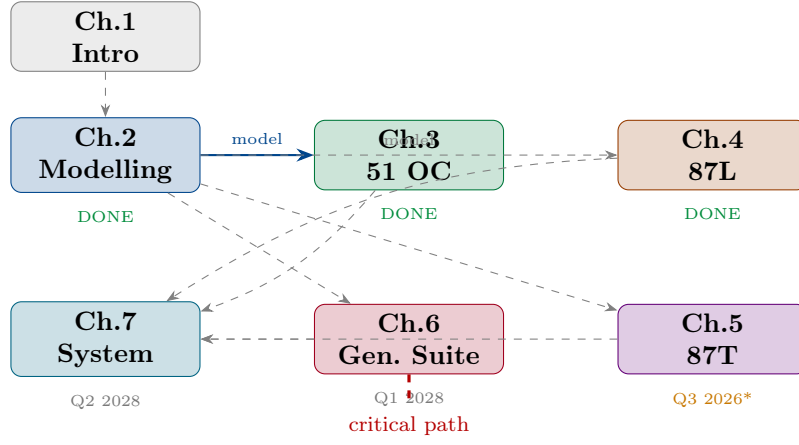


Figure 1: Chapter dependency graph. Solid arrows = strong prerequisite (model used directly). Dashed = logical dependency (motivation chain). Ch.6 (generator suite) is the critical path: it gates Ch.7.

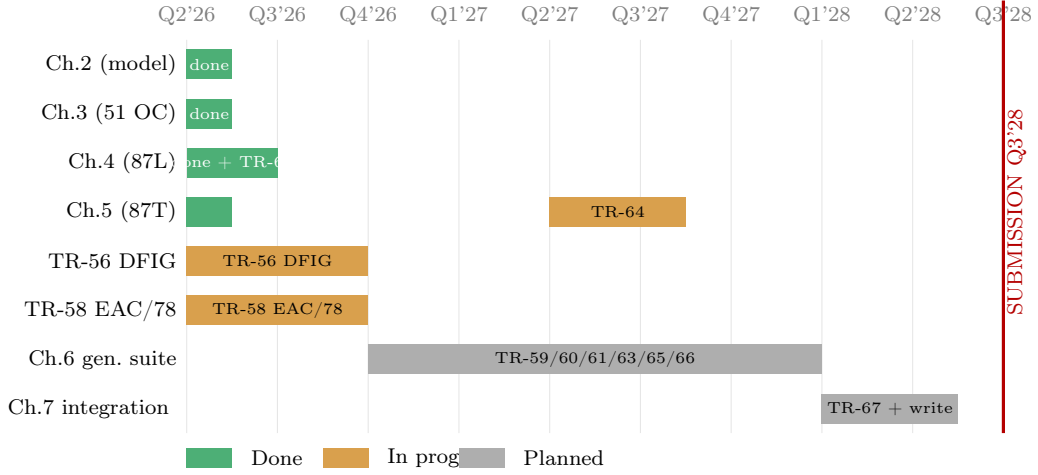


Figure 2: Thesis Gantt chart. Critical path runs through TR-56/58 (Q2–Q3 2026) to the full generator suite (Q4 2027) and HIL validation (Q1 2028).

7 Gantt Timeline

8 Rules for New TRs

Before starting any new TR (starting with TR-56), the following must be declared:

1. **Chapter home.** Which chapter does this TR feed? A TR that doesn't fit any chapter does not get written.
2. **Section slot.** Which §X.Y within that chapter? The chapter outline (Sections 3) defines all section slots. A new TR must either fill an existing slot or explicitly request a new slot (which requires justifying why the chapter outline was incomplete).
3. **Unifying problem contribution.** One sentence: what failure mode does this TR fix, and in which operating condition? Must be distinct from all existing TRs.
4. **Evidence standard.** Minimum: one simulation validation study (deterministic or MC). Maximum: PSCAD + HIL (deferred to TR-67).
5. **Journal target.** Which venue? A TR without a journal target is an internal document only and does not count toward the publication tally.

TR-56 declaration (applying these rules):

- Chapter: 6 (Generator Suite)
- Section slot: §6.1 (DFIG machine model for protection)
- Unifying problem: DFIG fault current has variable-amplitude slip-frequency DC that the 4-param SG model cannot fit — the 10-parameter piecewise ODE model is needed before any DFIG protection element can be designed
- Evidence: deterministic 10-scenario study + CUSUM crowbar detection test
- Journal: IEEE TEC

9 What This Map Does Not Include

The following topics are explicitly **out of scope** and must not be added to any chapter without a scope change review:

- Voltage ride-through (LVRT/HVRT) control — a generation problem, not protection
- Unit protection of PV strings or BESS internal faults — covered by IEEE 1547
- Cybersecurity of protection communications — not a relay-level problem
- Machine learning (ML) for protection — ruled out at thesis scoping stage (ML models have no IEC/IEEE standards path; physics-based estimators do)
- Market/economic optimisation of protection settings — different discipline

10 Summary Statistics

Table 4: Thesis Statistics at Lock Date (April 2026)

Metric	Complete	Planned
Technical reports (TRs)	55 + 2 (TR-57, 62) = 57	+10 (TR-56–67 minus done)
Journal papers complete	4 (A, B, D, E) + partial (C, F)	+4 planned
Python modules (protection)	12+ (line_diff, sync_oc suites)	+6–8 (generator suite)
Thesis pages written	~100 (Ch.3–4 + background)	+125 (Ch.5–7)
Published/submitted papers	0 (none submitted yet)	6 journal targets
MC trial count (to date)	15k (87L) + 5k (SPRT plan)	+35k (generator suite)